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Development of a Computer Program for the Dimensioning of Hydrofoil Boats with Fully Submerged Foils - *HYDROFOIL BOAT*

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Abstract

This work presents the development of a computer program for the dimensioning of fully submerged foils for hydrofoil boats, from a pre designed suitable high speed craft. The program is part of a design system named Solution Focused Design, also part of a subroutine called Dynamic Equilibrium, for the hydrodynamic design of planing boats and other high performance craft.

The program's routines were coded to apply the method created by Calkins and developed by Schachter and Vianna (1996), shown in this work, and uses several references of hydrofoils, such as Du Cane, Calkins, etc. From the high speed boat particulars, selected Seastate, NACA profile (or a customized foil), foil material, configurations (types such as U, inverted T or Pi), layouts, the routines draw the aft and forward configurations to scale. Soon after the program determines the foils' areas, chords, angles of attack, lift (L, with corrections for aspect ratio, free surface effect) and drag (D, induced, frictional and viscous pressure, wavemaking) forces, subject to cavitation and structural strength restrictions, solved for the thickness-chord ratio (t/c), for a range of load factors (weight-area ratio, W/S), to find the optimum foils dimensions and angles of attack, aft and forward.

Several graphs are presented by the program: t vs W/S for Maximum L/D, Cavitation and Structural Strength limits, Lift and Drag vs Speed, Angle of Attack vs Speed, Take-off Speed determination graph (Lift-Weight vs Speed), Resistance and Power vs Speed. As it can be seen, this program is aimed to help design any hydrofoil boat. A design example is shown.

1. Introduction

This work presents the development of a software (Fonteles, 2019) to dimension foils of a hydrofoil boat, using several references and formulation on hydrofoils, applying a method created by Calkins and published by Schachter and Vianna (1996), in this case for fully submerged foils. The program also deals with surface piercing foils, but this part is not presented in this work. This program is part of the suite named Dynamic Equilibrium (DynEq or EqDin), in turn part of a computational system under development, the Solution-Focused Design System (SFDS). The suite EqDin (DynEq), of which this program is part of, deals with the dynamic

equilibrium (lift, drag, balance) of planing and hydrofoil boats, as well as other high performance vessels, that also allows for propeller selection.

SFDS (Souza, 2014) is a system based on a design philosophy and methodology proposed by Schachter et al, 2006, as a development from Schachter 1990 and Keane et al, 1991. It is integrated, specially encoded to allow for processing conflicting naval architecture design modules out of a usual sequence, thus enabling to creatively test different design options, focused on the solution.

The program of this work, part of the EqDin suite, named *Hydrofoil Boat*, works both integrated to the system and autonomously, as a single

software. As mentioned, it applies the cited method, that from a stipulated foil configuration, from a pre-defined high-speed hull form, dimensions interactively the foils, calculating Lift and Drag, subject to constraints such as Structural Resistance and Cavitation, as a function of thickness to chord (t/c) and load factor (W/S). It is an absolute novel computer program, since all others in existence compute either theoretically or with CFD the foils alone or interacting with a hull, only as given, but never in a design integrated and interactive fashion, it applies a hydrofoil boat design methodology, with foil dimensioning for equilibrium, take-off, angle of attack determination and variation, etc. Also, due to the design multidisciplinary approach mentioned, for consistency, the formulation adopted is analytical empirical, experimental based.

2. The Design Method and Formulation

As mentioned, the design method of this program for dimensioning the foils for a hydrofoil boat were created by Calkins (1976), uses several formulations, mainly by Du Cane (1973), and was presented by Schachter and Vianna (1996).

The method was conceived to optimize the main features of foil dimensioning design, which are: Lift and Drag to, respectively, counteract the weight and for the propulsion, and the ratio of Lift on Drag coefficients, C_L/C_D , should be maximized, subject to structural strength and cavitation limits constraints. These parameters were all solved for two important hydrofoil design factors, thickness on chord ratio (t/c) and a load coefficient (W/S , weight on foil area), so they could be brought together in the optimization graph of Figure 1.

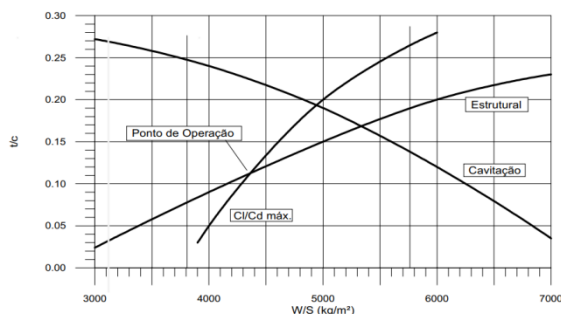


Figure 1 – Features Limits and Operation Point

This method shown in the graph was created to use a generic, self-designed foil section, and therefore allowing the freedom to vary the foil thickness, t . For the most common cases, where tested and tabled profiles, such as NACA, p. e., are

used, t/c is constant, optimized for a maximum C_L/C_D , thus represented by a straight line in the graph (Figure 2). Another fundamental design factor to be considered is the load coefficient W/S . This is a ratio taken from experience from hydrofoil boats, with values of practical limits between the foils leaving the water and the boat being too heavy to take-off. These practical values of W/S vary from 3800 to 5700 kgf/m^2 (800 to 1200 PSI), and the program uses the range 3000 to 7000 kgf/m^2 for better curve adjustments and visualization.

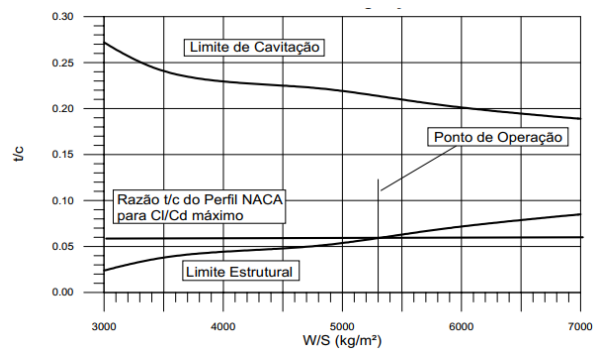


Figure 2 – Real case situation, using a NACA foil

There are, of course, other design factors to be considered in a concept design of a hydrofoil boat, such as stability, structural design, propulsion, etc., but these are out of the scope of this program.

The foils are dimensioned to work up to the cruise speed, with the take-off speed determination, showing several graphs of all results.

The lift of the foils is varied by the speed and their submersion, therefore the lift is calculated at each knot along the take-off process. The take-off speed is defined in the program by the subtraction of the lift of the foils from the weight, until the speed the lift equals the weight (Figure 3). The displacement at each knot is used to compute the resistance of the hull (Holtrop 1984 up to $Fn=0.4$ and Savitsky 1964 above), which is added to the foil drag to provide the Resistance/Power vs Speed curve.

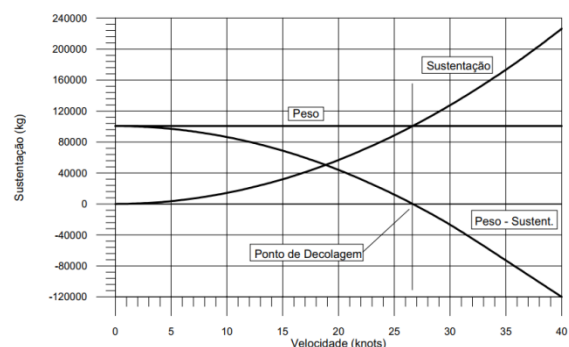


Figure 3 – Take-off speed determination

2.1. Foil Lift (fully submerged)

The foil lift (L) is given by:

$$L = \frac{1}{2} \rho V^2 S C_L \text{ (kgf)} \quad (1)$$

Where ρ is the specific gravity of the fluid ($\text{kgf s}^2/\text{m}^4$), S is the foil area (m^2), as the span (b) times the chord (c) of the foil.

The lift coefficient (C_L) is defined as:

$$C_L = C_{L\alpha} \alpha_T \quad (2)$$

Where $C_{L\alpha}$ is the slope of the lift curve and α_T is the total angle of attack (Figure 4). The lift coefficient C_L in normal conditions varies linearly with the total angle of attack (α_T) and has a slope ($C_{L\alpha}$) value of 2π , for two dimensional thin foils. NACA provides tabled C_L for their foils.

$$C_L = 2\pi \alpha_T \quad (3)$$

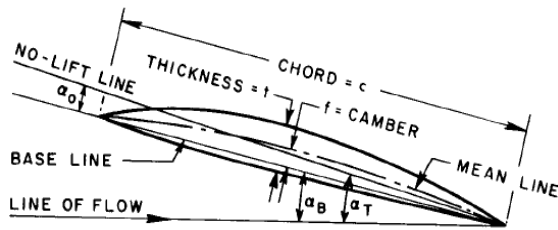


Figure 4 – Hydrofoil section geometry (Du Cane 1973)

The correction for the angular coefficient, $C_{L\alpha}$, of the lift curve for fully submerged foils taken to a reasonable practical minimum (to interact with drag components) are a correction for the Aspect Ratio (AR), proposed by Calkins (1976) and applied successfully by Schachter (1978);

$$AR_{\text{corrected}} = AR / AR + 2 \quad (4)$$

and a correction for the Free Surface Effect (K), by Wadlin, presented by Du Cane (1973), considering the foil submersion (h_f) and the chord (c):

$$K = [(4 h_f / c)^2 + 1] / [(4 h_f / c)^2 + 2] \quad (5)$$

Leading to the correction of the angular coefficient of the lift curve as:

$$C_{L\alpha} = 2\pi (AR / AR + 2) [(4 h_f / c)^2 + 1] / [(4 h_f / c)^2 + 2] \quad (6)$$

2.2. Foil Drag (fully submerged)

The viscosity of a real fluid has a negligible effect on the lift of thin foils, but not on the drag. The drag (D) calculation is analogous to the lift.

$$D = \frac{1}{2} \rho V^2 S C_D \text{ (kgf)} \quad (7)$$

The drag coefficient C_D , for fully submerged foils has 4 main components, frictional and viscous pressure (C_{D0}), wavemaking, C_{DW} and induced drag C_{Di} .

$$C_D = C_{D0} + C_{DW} + C_{Di} \quad (8)$$

The formulation is given by Du Cane (1962). The coefficient C_f is from the ATTC flat plate friction table, including the known correction for high-speed craft ΔC_f , which in the case of foils is 0.0008:

$$C_{D0} = 2(C_f + \Delta C_f) (1 + 1.2 t/c) + 0.11 (C_{L0} - C_L)^2 \quad (9)$$

The wavemaking component, presented by Du Cane, developed by Wladimirov, is a function of $C_{L\alpha}$, the inclination of the lift coefficient curve (as all drag), the chord's Froude number (Fn_c) and the foil submersion (h_f) and chord (c):

$$C_{DW} = (C_{L\alpha})^2 \frac{1}{2Fn_c^2} \text{EXP}[-2(h_f/c) Fn_c^2] \quad (10)$$

The lift induced drag can be estimated with the realistic supposition of elliptic transversal loading, as formulated by Du Cane (1962) and Hoerner (1965). The induced component can be calculated as a function of $C_{L\alpha}$, biplane factor (σ) and a corrected aspect ratio (AR_c):

$$C_{Di} = (C_{L\alpha})^2 / (1 + \sigma / (\pi AR_c)) \quad (11)$$

The biplane factor (σ) for Prandtl wings of finite span is expressed by Du Cane. It is nil for wings in the air, but significant in the water, and although of complex determination, was simplified by Eames for high Froude numbers, such as in hydrofoil boats:

$$\sigma = AR_c / [AR_c + 12 (h_f/c)] \quad (12)$$

$$AR_c = b/c [1 + (a/b)^3 h_f/b] \quad (\text{see Figure 5}) \quad (13)$$

2.3. Foil Angle of Attack (α_T)

From equations (1) and (2),

$$\alpha_T = C_L / C_{L\alpha} \quad (14)$$

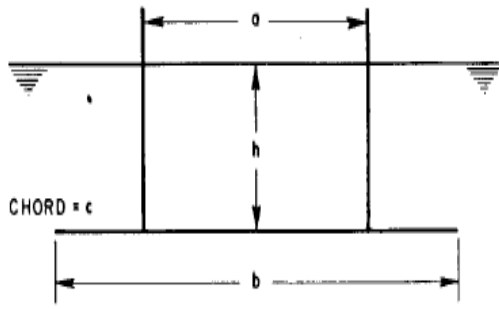


Figure 5 – Foil configuration, varying from inverted T through π , up to U shape, as 'a' increases. ($h = h_f$)

2.3. Structural Limit (fully submerged)

For the structural strength constraint limit, the bending moment was used, since it is more critical than the shear force. A formulation was developed to obtain a minimum t/c , as a function of W/S that satisfied this constraint.

This formulation can be used for inverted T, inverted π and U foil configurations. The inverted T deduction will be presented here, the other are analogous. The load distribution in an inverted T foil is:

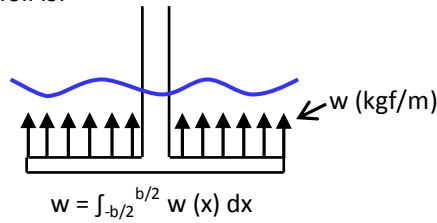


Figure 6 – Loads on an inverted T foil

That can be considered as studded (cantilever) beams:

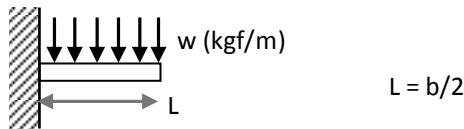


Figure 7 – Studded beam

And the Moment can be written as:

$$M_{(\max)} = w L^2 / 2 = w b^2 / 8 \quad (15)$$

The inertia (I) was taken from a generic flat-ogival foil, which is a reasonable approximation for any foil.

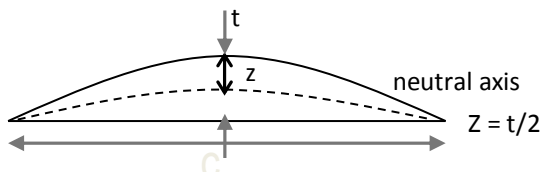


Figure 8 – Flat-ogival foil

$$I = \pi c t^3 / 64 \quad (16)$$

And the Yield Stress (σ_y) is,

$$\sigma_y = M_{(\max)} z / I \quad (\text{the Section Modulus being } z/I) = [(w b^2 / 8) t/2] / (64 / \pi c t^3) \quad (17)$$

Creating a constant with the numbers and π , and solving for t/c , considering that $W/S = wb/bc$, the formulation becomes:

$$t/c = \sqrt{[(1.27 W/S b^2) / (\sigma_y c^2)]} \quad (18)$$

Working a general equation from studded (T) to bi-anchored (U) beams, with a correction for strut spacing (b_s) from 0 to b and W/S in IS units (SF: safety factor):

$$t/c = \sqrt{0.85 b_s^2 + 2 \{1.27 [(b - b_s)/2]^2 g(W/S)\} / (\sigma_y c^2 / SF)} \quad (19)$$

2.4 Cavitation Limit

For the constraint of the cavitation limit, the method works out a formulation to obtain a t/c , as a function of W/S , where the foil does not cavitate. For such, a practical and approximated formulation, proposed by Du Cane, was used, that provides an incipient cavitation number (σ_i), as:

$$\sigma_i = A_f t/c + B_f C_L \quad (20)$$

This formulation assumes that the total pressure distribution is approximately the summation of individual contributions, it is for thin foils ($t/c < 0.2$) and $\alpha_T = 0$. 'A_f' is a practical profile factor (see Table 1).

$$B_f = 1 / 1 + a \quad (21)$$

Where 'a' is the camber or taper (mean line) height, as 'f' in Figure 4. 'B_f' varies from 0.5 to 2/ π .

Table 1 – Profile Factors A_f

Ellipse (Ideal but not practical)	$\sigma_i = 2.00 t/c$
D.T.M.B.-E.P.H. (Submerged struts only)	2.15 t/c
N.A.C.A. 16 Series	2.28 t/c
Warren 45°	2.36 t/c
N.A.C.A. 66 Series	2.42 t/c
Circular Arc (Bi-ogival)	2.55 t/c
N.A.C.A. 65 Series	2.58 t/c
N.A.C.A. 64 Series	2.65 t/c
N.A.C.A. 63 Series	2.67 t/c
N.A.C.A. 4 and 5 Digit. (For $t/c > 0.08$)	3.50 t/c

From equation (1), C_L is:

$$C_L = L / \frac{1}{2} \rho V^2 S = W/S / q \quad (22)$$

Where L (lift) = W (weight); $q = \frac{1}{2} \rho V^2$

$$\text{Hence, } \sigma_i = A_f t/c + B_f (W/S)/q \quad (23)$$

The incipient cavitation number also is the atmospheric pressure (p_a) minus the vapor pressure (p_v) plus the hydrostatic pressure ($\rho g h_i$), over q . Equating equation (18) to this expression, attributing values for p_a , p_v and ρ at 15 deg C, for $g = 9.81 \text{ m/s}^2$, and solving for t/c , the expression of the maximum t/c to avoid cavitation becomes:

$$t/c = [69.06 + 10.045 h_i - B_f (W/S)] / A_f q \quad (24)$$

2.5. Determination of $(C_L/C_D)_{\max}$

For the design using a generic foil, the maximum Lift to Drag Coefficient Ratio, $(C_L/C_D)_{\max}$ is obtained from equations (22), (8), (9) and (10):

$$(C_L/C_D)_{\max} = [W/S / q] / (C_{D0} + C_{DW} + C_{DI}) \quad (25)$$

For a tabled foil section, such as NACA, the t/c found in the design process is already for a $(C_L/C_D)_{\max}$.

3. Design Sequence

As mentioned, this program dimensions the foils of a pre-defined high-speed hull, adequate for being a hydrofoil boat. This can be done either in the Solution-Focused Design System this software is integrated with, or by a hull defined with other software (stand-alone mode).

The Hydrofoil Boat program, as part of the DynEq (EqDin) suite, uses for resistance (or drag) calculations, in both cases, and automatically in the suite, the Holtrop method when Froude numbers are below 0.4 and the Savitsky method for Froude numbers of 0.4 onwards.

The program is processed interactively, in the following sequence:

- 1) The parameters required can be obtained either directly from a '.eqd' file or as user input: design displacement (Δ), center of gravity (LCG, VCG), cruise speed (V), distance from propeller thrust (T) to VCG, length between perpendiculars (L_{pp}), breadth (B_{OA}) and breadth at chine (B_c), depth (D), deadrise angle (β), shaft inclination (ϵ), chine height at transom, strut height margin, foil structural

strength safety factor (SF) and the boat's Hydrostatics table (from *Free!Ship* software).

- 2) In the 'Wave Height' area the user defines the design operating Seastate.
- 3) In the foil definition area, for the Rear and Front Foils, the user selects the Foil Profile, its Material, Configuration (inverted T or π , U or V surface piercing), the longitudinal position of each foil (for the determination of the load on the foils) and the surface roughness.
- 4) In the Foil Geometry area the design starts. The user inputs both foils geometries (spans, strut spans, inclinations), and their feasibility or not is shown in real time for modifications.
- 5) By clicking in 'Calculate', the computed information of the foils can be verified in the Rear and Front Foil Definition area (areas, chords, Lift, angles of attack, Structural Strength, Cavitation, for a range of W/S), pointing out the optimized foils.
- 6) With the foils defined, the next step is the take-off procedure, which allows for two options: either prescribing the maximum foil angle of attack (for the earliest take-off) or the desired take-off speed. It is a graphical procedure with lift of the foils vs decreased draft of the hull. The design criteria for angles of attack of this procedure is to use a constant angle of attack until take-off, and then decrease it at each knot in order to keep the lift (equal to the weight) constant until the cruise speed is reached. Graphs of lift, drag, angle of attack vs speed and other are presented.
- 7) From the imported Hydrostatics the Resistance (Drag) and Power vs Speed graphs are shown.

4. Description and Utilization of the Program

The *Hydrofoil Boat* program is a part of the EqDin (DynEq) suite and is fully integrated, using its subroutines. It is also integrated with the Solution-Focused Design System, through files that interleave to and from each other. The program follows the System's same philosophy: a minimum of screens, a logical sequence of required data, self-explained (names and units) and all necessary data (input and output) on the same screens and ease to change from one screen to another, with consistency of data on modifications. The main screen can be seen in Figure 9.

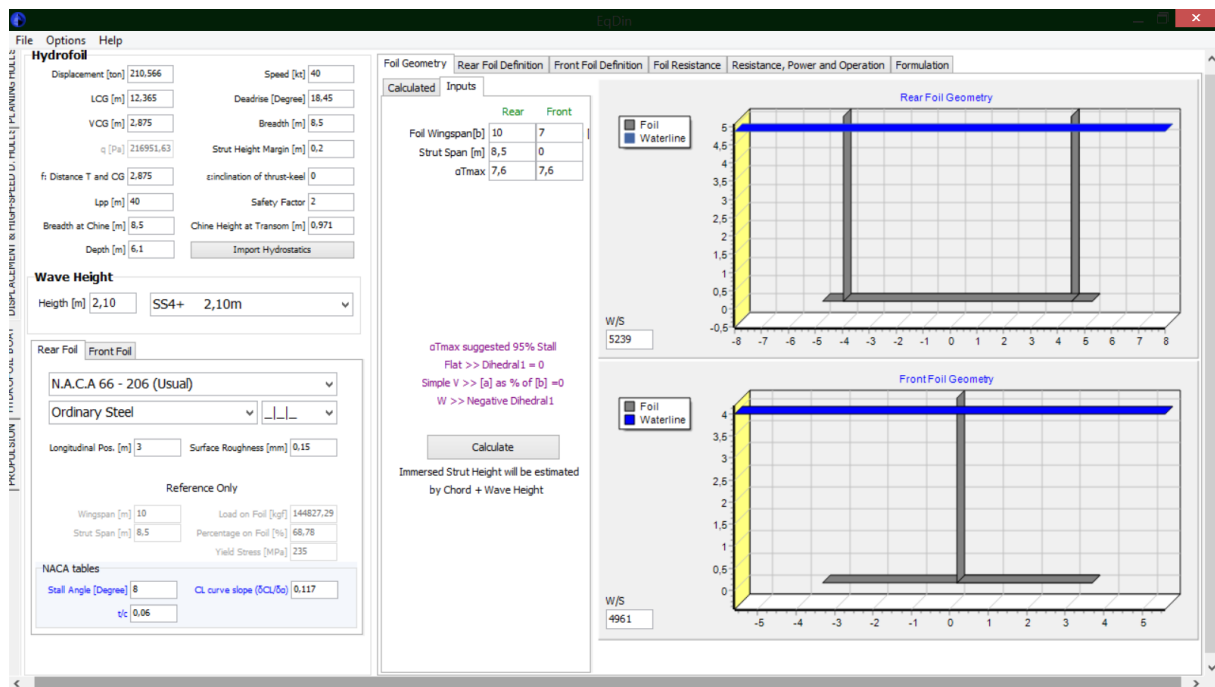


Figure 9 - Hydrofoil Boat's main screen

The hull references, as in the whole SFD System, are the stern, the baseline and the centerline, for all data. Besides the design functions of the program, there are the general commands of the whole suite, which are consistent with the other programs of the System.

4.1. General Commands

These are interface functions of DynEq (EqDin) such as 'File', 'Options' and 'Help'. In 'File' (Figure 10), where:

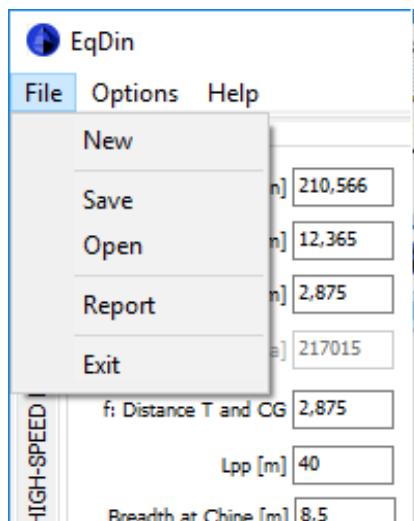


Figure 10 – File Interface

New – erases all input and output data;

Save – saves the actual state of the design process;
Open – opens a file generated by the program;
Report – generates a report of the results (Fig. 11);
Exit – closes the program.

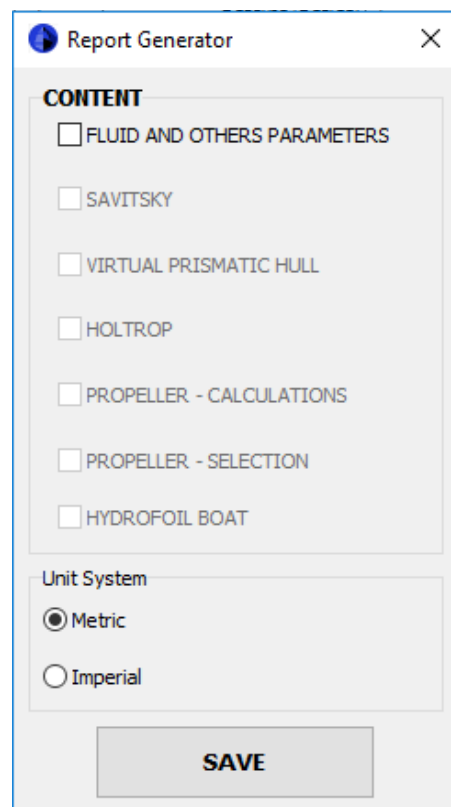


Figure 11 – Report Generator Interface

In the 'Options' interface (Figure 12), there is:

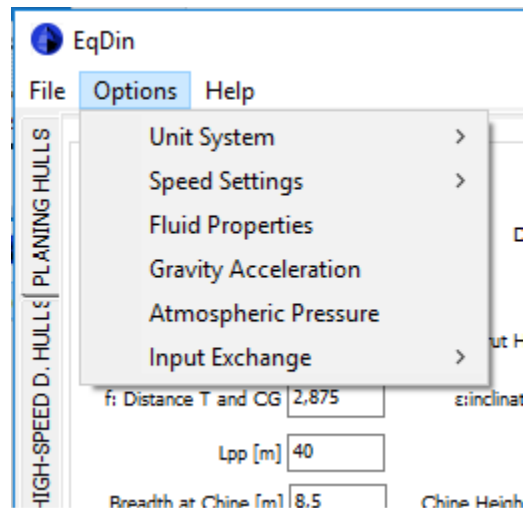


Figure 12 – Options Interface

Unit System – used to determine the units to be used in the calculations. This option is applied to many programs of EeqDin, but *Hydrofoil Boat* is limited to the metric units;

Speed Settings – to define several speeds and ranges for the programs *Planing Hulls* and *Displacement and High-Speed Displacement Hulls* and it is not used by the *Hydrofoil Boat* program;

Fluid Properties – the designer may indicate the density and viscosity of the fluid considered, as well as its saturation pressure (in both units). To ease the application, the values were parameterized with the temperature the user can prescribe, both for salt or fresh water (the retrievable default being salt water at 20 degrees Celsius);

Gravity Acceleration – input option for the value of gravity acceleration. The retrievable default is 9.81 m/s^2 | 32.19 ft/s^2 ;

Atmospheric Pressure – input option for the value of atmospheric pressure. The retrievable default is 101.32 kPa | 14.70 Psi ;

Input Exchange – is a tool implemented for data consistency between calculation methods for *Planing Hulls*, *Displacement* and *High-Speed Displacement Hulls* and *Propulsion*, to allow for the interleaving of input data from one to another, not used by this program.

4.2. Main Area of the Program

As it can be seen in Figure 15, the screens were designed to create interfaces that follow the system's philosophy of concentrating the maximum possible information in a minimum number of screens, with a logical input sequence, either separating inputs from output when starting

with general data, and logically allowing for inputs and their local outputs within a calculation subject. In Figure 15, the area **A** is for initial general input, in a logical design sequence, and the **B** area for the output, but allowing for local adjustments and more detailed inputs for the foils design, enabling for interactions, by trial and error, on the drawing of the foils (span size, configuration, inclinations), for fully submerged or surface piercing foils.

4.2.1. Data Entry Area (A)

In this area the general input of the program is entered, either by an 'eqd' (for most data) file of the suite, or by hand. The first group is called **Hydrofoil** (Figure 13) and requires the data mentioned in Section 3. 1).

Figure 13 – Data entry **A** – Hydrofoil group

The second group is called **Wave Height** and it is where the user inputs the wave height or selects the operating design Seastate from the Pierson-Moskowitz spectrum (usual for high-speed craft).

Figure 14 - Data entry **A** – Wave Height group

The third group (Figure 16) is for the initial dimensioning of the **Rear Foil** and **Front Foil**: the user defines the profile, configuration (that may be changed interactively), material, longitudinal position on the boat (referenced to the stern), surface roughness.

For the selection of the profile (Figure 16a), the user can chose either one of four NACA profiles provided (for which their Stall Angles, t/c , C_L curve

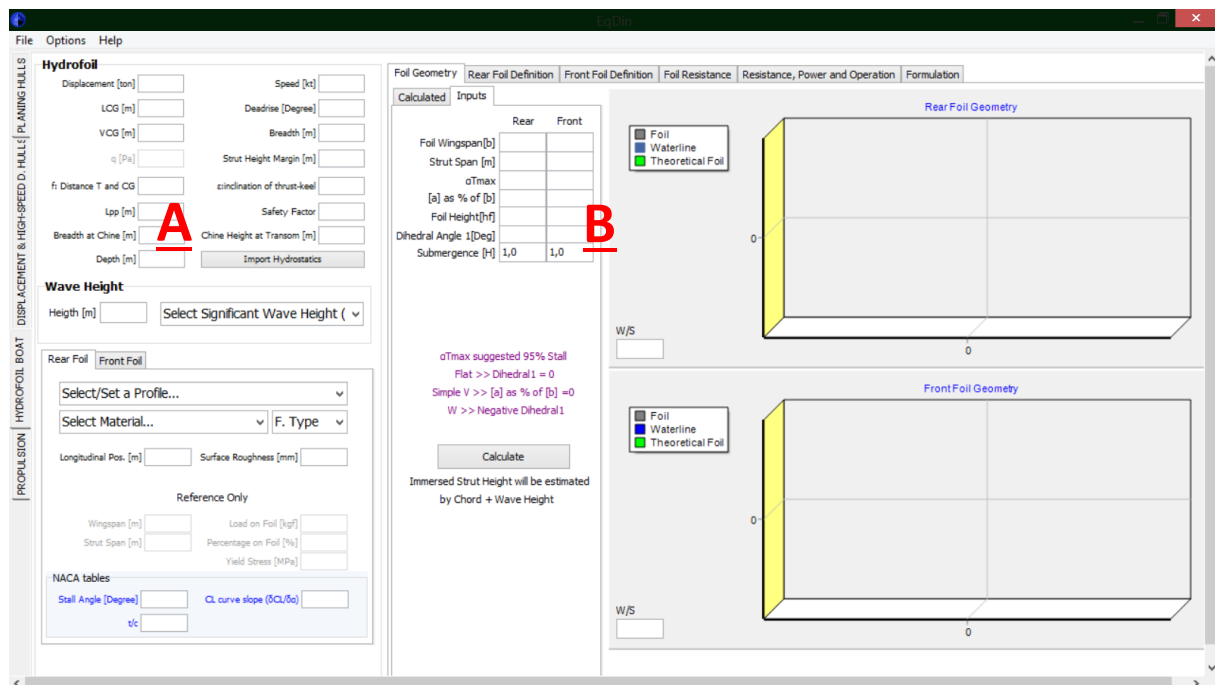


Figure 15 – Main Area of the Program

slope are given), or his own profile (Custom) and if it is a NACA section, these values must be given by the user.

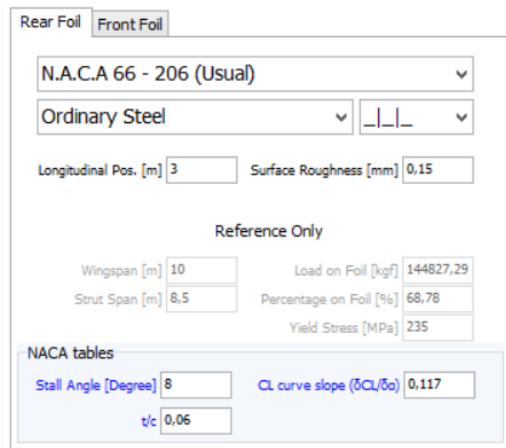


Figure 16 – Data Entry A - Rear Foil and Front Foil

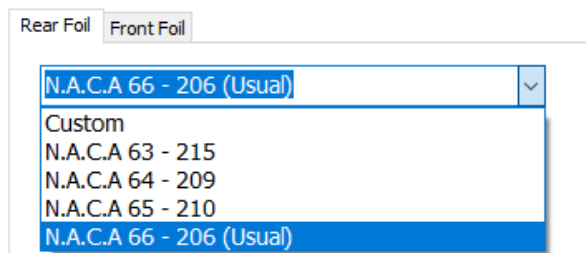


Figure 16a – Section Profile Selection

The Configuration selection can be seen in Figure 16b. Here the user decides the design configuration of the Rear and Front foils, that can be inverted T or π , U (fully submerged) or V,

double V, W, flat bottom V (surface piercing/ dihedral).

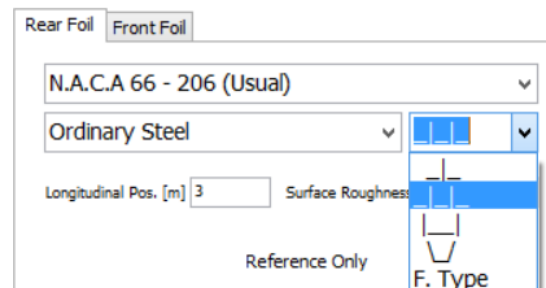


Figure 16b – Foil Configuration Selection

The foil material, for the structural strength constraint calculation, can be selected by the user as in Figure 16c.

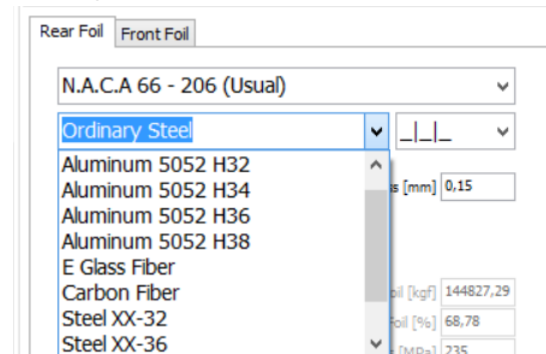


Figure 16c – Foil Material Selection

Finally, the Longitudinal Position of each foil must be prescribed by the user, for the loads on each foil and the take-off calculations, and so the Surface Roughness (Figure 16).

4.2.2. Design and Results Area (B)

In this area (B in Figure 15) there are five groups: Foil Geometry, Rear and Front Foil Definition, Foil Resistance, Resistance, Power and Operation and Equations. In Equations, the last group, there is a display of the formulation shown in Section 2.

4.2.2.1 Foil Geometry Area

The Foil Geometry area is where the foils are dimensioned, completing the Data Entry A inputs, with the foil wingspan, the strut span (see Strut Spans and drawings in Figure 9) and the maximum angle of attack (based on the stall angle). The parameters given at this point are related to the geometry of the foils, functions of the breadth, and are immediately drawn as data is typed. There is a real time pre-calculation to point out if the

parameters given are feasible or not, by turning the words 'Rear' and 'Front' of each foil column green or red (see Figure 9). The process is highly interactive, and the user can change at any time the foil geometry and even any parameters from the Data Entry. There are other settings for the type configuration and drawing of surface piercing foils, but are out of the scope of this work.

4.2.2.2 Rear and Front Foils Definition

When the user presses 'Calculate', all the calculations of the method are done and tabled (Figure 17 is an example for the rear foil), as can be seen in Rear and Front Foils Definition, and the charts optimizing both Foils are drawn for t/c as a function of W/S, subject to Structural Strength and Cavitation constraints (Figure 18, example for the rear foil).

Foil Geometry		Rear Foil Definition		Front Foil Definition		Foil Resistance		Resistance, Power and Operation		Formulation	
		Chart		Table							
Lift [N]	1422459,37	W/S[kg/m²]	5100	5150	5200	5250	5300	5350	5400		
Chord [m]	2,765	Area[m²]	28,398	28,122	27,851	27,586	27,326	27,071	26,82		
Cl Alfa	0,081	Chord[m]	2,84	2,812	2,785	2,759	2,733	2,707	2,682		
Alfa [Degree]	2,928	ImmStrut Ht[m]	4,94	4,912	4,885	4,859	4,833	4,807	4,782		
W/S [kg/m²]	5239	Aspect Ratio	3,521	3,556	3,591	3,625	3,659	3,694	3,729		
W/S Pratical Limits 3800-5700 kg/m²		Calkins Corr.	0,638	0,64	0,642	0,644	0,647	0,649	0,651		
		Wadlin Corr.	0,98	0,98	0,98	0,981	0,981	0,981	0,981		
		Corrected AR	4,589	4,629	4,668	4,707	4,745	4,785	4,824		
		Cl alpha	0,08	0,08	0,08	0,081	0,081	0,081	0,081		
		Cl required	0,231	0,233	0,235	0,237	0,24	0,242	0,244		
		Ang. Att	2,888	2,913	2,937	2,926	2,963	2,988	3,012		
		Cav. Lim.[t/c]	0,2501	0,249	0,248	0,2469	0,2459	0,2449	0,2439		
		Stru. Lim.[t/c]	0,0576	0,0585	0,0593	0,0602	0,061	0,0619	0,0628		
		CL/CD max[t/c]	0,06	0,06	0,06	0,06	0,06	0,06	0,06		

Figure 17 – Rear Foil Definition Table (where: Cav. Lim. – cavitation limit; Stru. Lim. – structural limit)

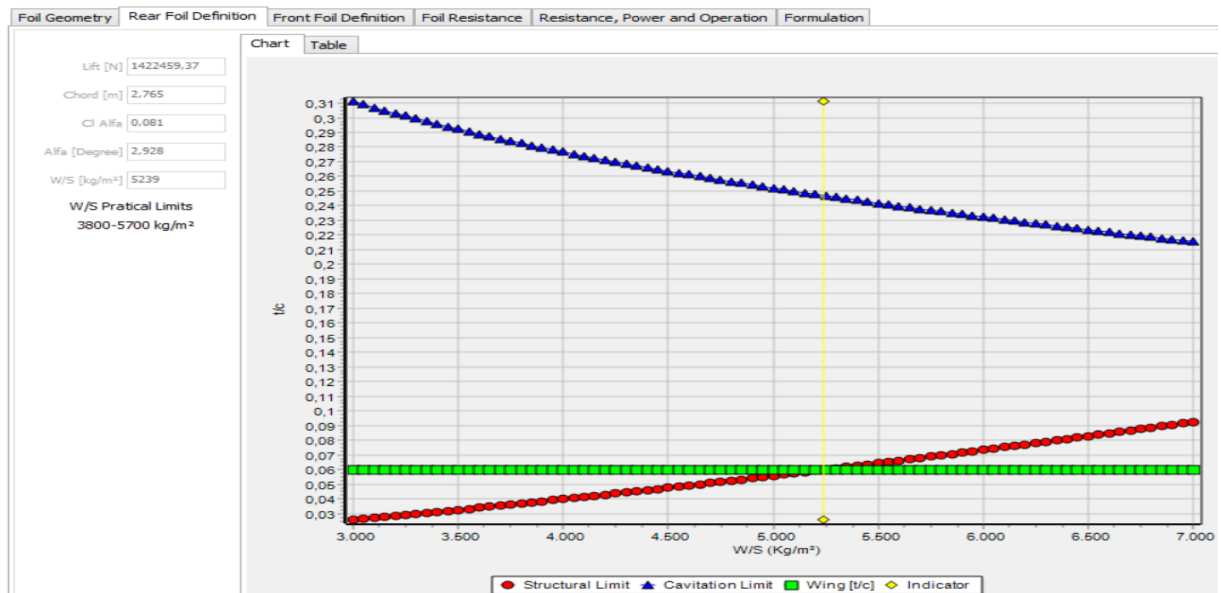


Figure 18 – Rear Foil Optimization Chart

4.2.2.3 Foil Resistance Area

This is the area where the **Take-off Speed** is found by calculating progressively the foils Lift, at each knot, until the weight of the hydrofoil boat is reached. The boat's displacement is the subtraction of the foil's lift from the boat's weight at each knot. Each of these boat displacements is used to calculate the boat's resistance (for Froude Numbers less than 0.4, by Holtrop) and the boat's drag (by Savitsky, Froude 0.4 onwards). This can be seen in the graph of Figure 19.

There are two options to obtain **Take-off**, one by using the maximum possible, pre-stall, angle of attack (named Initial Angle) of the selected foil to find the speed, and the other by prescribing (a

reasonable) take-off speed to find the angle of attack. This setting is on the top left of Figure 19. On the bottom left the pre-take-off condition can be seen (values independent of speed).

As mentioned, the design criteria used is to use a fixed angle of attack up to the take-off and then keep lowering this angle at each speed until the cruise speed is reached, so that the boat doesn't bounce out of the water. This can be seen in Figure 20 together with the foils' lift and drag curves, and the cruise condition (top speed) is shown in the bottom left.

The **Lift** and **Drag** calculations of the Front and Rear foils are in the chart of Figure 21.

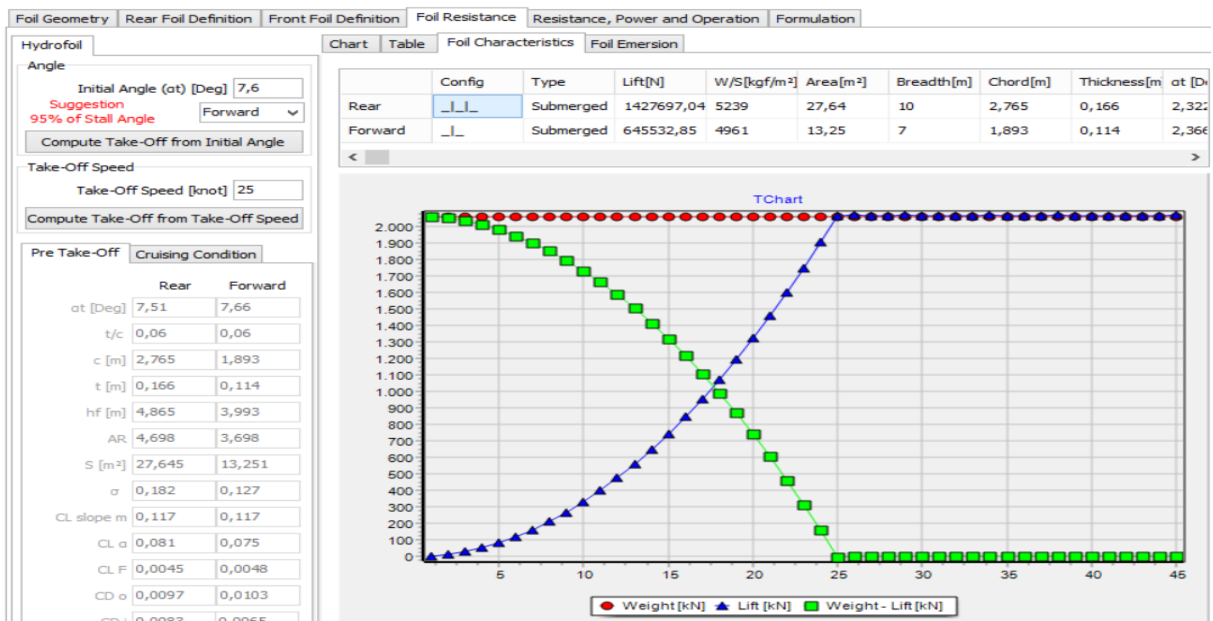


Figure 19 – Foil Resistance Area – Take-off Speed

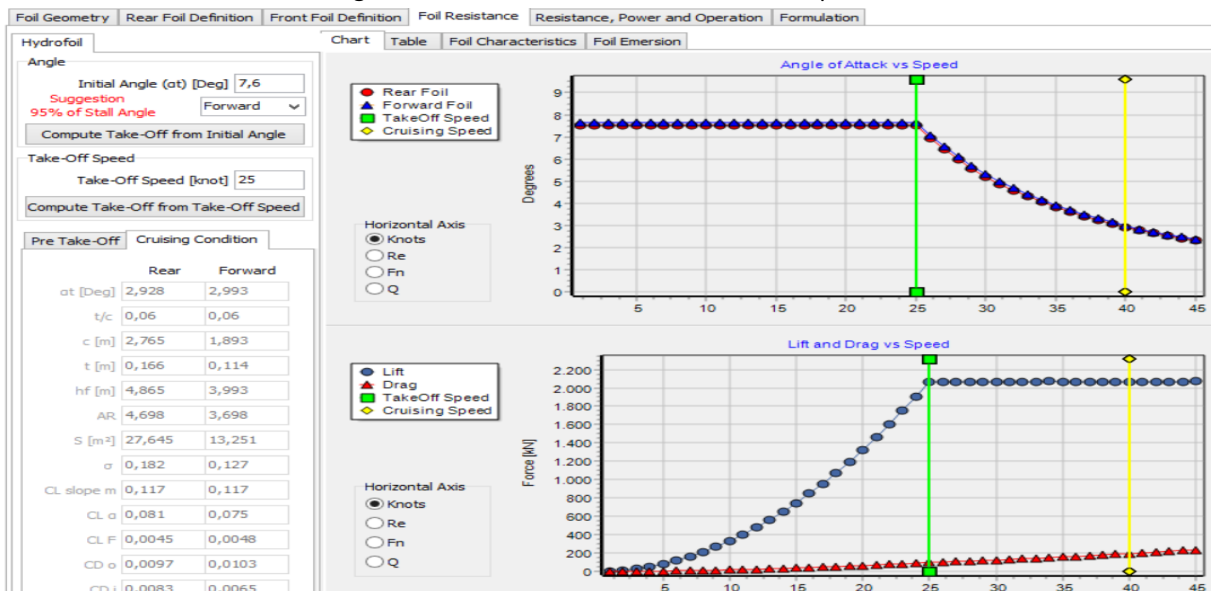


Figure 20 – Foil Resistance Area – Angles of Attack and Foils Lift & Drag Curves

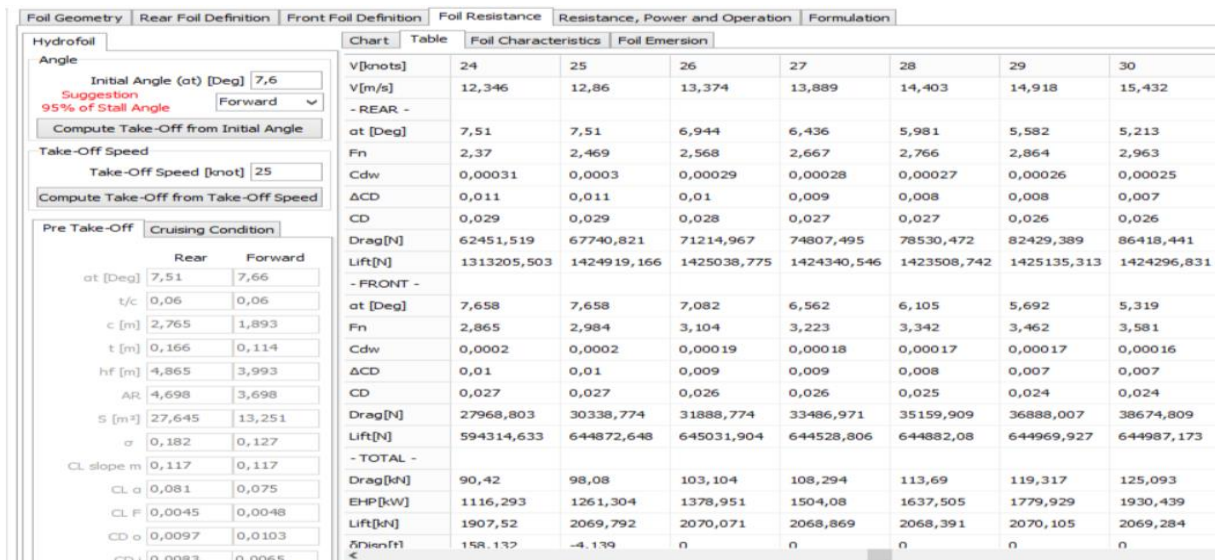


Figure 21 – Foil Resistance Area – Foil Lift & Drag Calculations Chart

There is a 'Foil Emersion' tab in this area that is exclusively for surface piercing foils, not part of this work.

4.2.2.4 Resistance, Power and Operation Area

This is the area where the total Resistance (and/or Drag) and Power Curves of the whole craft are displayed. These values are computed as in Figure 21, complemented in Figure 22 and displayed in Figure 23.

The Planing Boat shown is the one on which the foils were dimensioned, and it can be either given as entry data in the *Planing Boat* program of the EqDin (DynEq) suite or designed in it. It should be noted that the calculations add foil and hull resistance/drag at lower speeds, and as the hull leaves the water (weight minus foil lift) the hull resistance/drag decreases until it leaves the water

(take-off speed), when only the foils' resistance/drag prevails. In the second graph there is a comparison between the Effective Horsepower (EHP) in kW of the hydrofoil boat and the planing boat without the foils.

Nomenclature of Figure 22:

δDisp: displacement condition,

L_{WL}: waterline length,

Q: Taylor Number - $V(\text{kt})/V_L(\text{ft})$,

Fn: Froude Number - $V/\sqrt{gL}$,

Tf, Ta: draft forward and aft,

B_{WL}: waterline breadth,

Sw: hull wetted area,

LCB: center of buoyancy,

CM: midship section coefficient,

Cwp: waterplane area coefficient,

At: transom area,

Hull RT: displacement regime resistance,

Pl. Bt RT: planing boat drag.

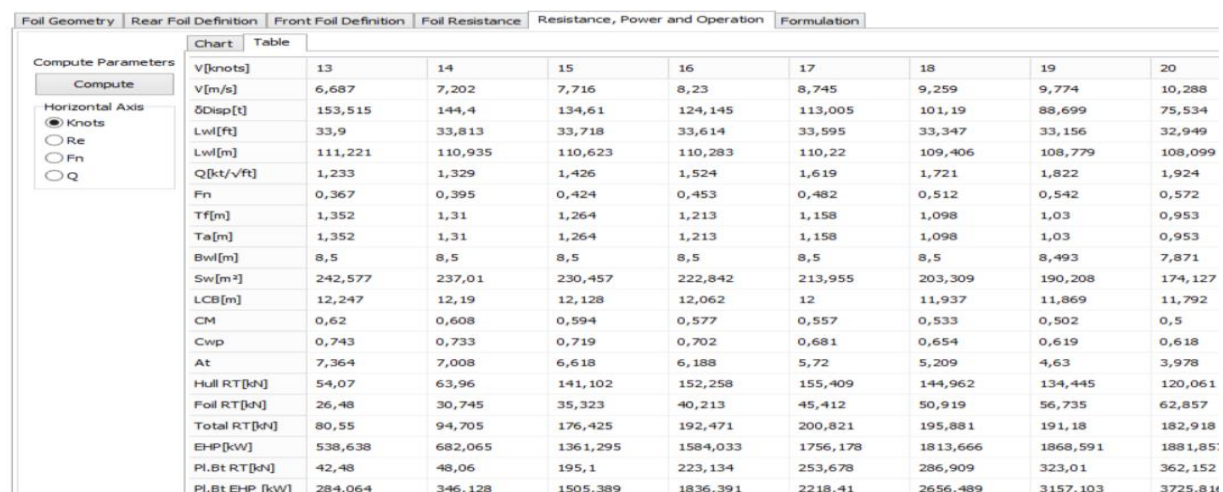


Figure 22 – Resistance, Power and Operation Area Calculations

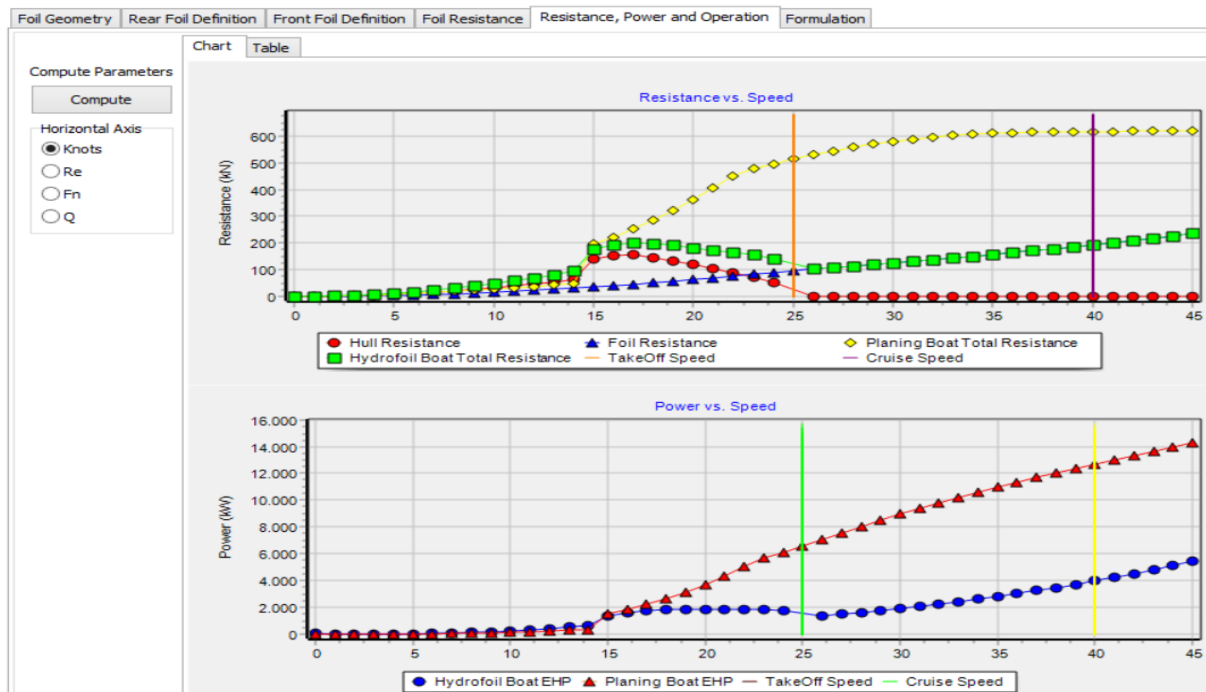


Figure 23 – Resistance, Power and Operation Area – Total Resistance and Effective Horsepower Comparisons

5. Precision Check

In order to verify the reliability of the program, few estimates based on real hydrofoil boats were made using the program to see if they gave the same results, as measured, as a precision check of the program calculations.

Information was gathered from datasheets and images of real existing hydrofoil boats, such as the Boeing JETFOIL 929 115 for fully submerged foils (Figure 24).

Some information was not available in the literature, such as the Section Profile, Angle of Attack at the calculated (cruise) speed and the position of the center of gravity of the boat. These parameters had to be estimated or worked backwards (for the LCG, p.e., from the displacement and the foils' positions).

For the JETFOIL 929 115, the following particulars were gathered:

$L_{OA} = 27.4$ m,

Breadth = 9.5 m,

Draft = 5.2 m,

Speed = 43 Kt

Displacement = 117 t,

VCG = 2.6 m.

LCG = 13.7 m

(Aft foil ~3 m and Fwd foil ~21.3m from the stern).

Fwd Span (b_f) = ~3.6 m

Aft Span (b_a) = ~9.5 m

Chord (c) = ~1.15 m (Fwd and Aft)

Estimated foil material: Steel,

Estimated Section Profile: NACA 63-215.

The comparison parameter was the calculated foil from the program.



Figure 24 – Boeing JETFOIL 929 115

Table 2 – JETFOIL: precision check

Foil	Measured Chord (c)	Calculated Chord (c)	Discrepancy
Aft	1.15	1.16	0.87%
Fwd	1.15	1.14	-0.87%

These discrepancies seem reasonable, as this is an initial dimensioning of the foils.

The JETFOIL uses an inverted T foil forward and a U foil aft. From the data and images used,

6. Design Example

A design example is presented here to demonstrate the method and the design process. It is a conceptual design of a fully submerged hydrofoil patrol boat, with the main particulars shown below, with the hull based on a planing boat developed in the *Planing Boat* program of the EqDin (DynEq) suite. This planing boat has $L_{pp} = 40$ m, $B = 8.5$ m and $D = 6.1$ m and the data is as shown in Figure 13:

Displacement (Δ) = 210.566 t,
 LCG = 12.365 m,
 VCG = 2.875 m,
 $f = 2.875$ m (thrust to VCG distance),
 Breadth at Chine = 9.0 m,
 Deadrise angle (β) = 18.45 deg (dominant),
 $\varepsilon = 0.0$ (shaft inclination angle – n.a.),
 Chine height at transom = 0.971 m.

This hydrofoil patrol boat was specified to have a 40 Kt cruise speed, to operate in Seastate 4 (Significant wave height ($h_{1/3}$) of 2.1 m), a Strut height margin of 0.2 was used, and a structural strength (SF) foil safety factor of 2.0 was specified. The Seastate was selected as in Figure 14. For the foils, the Rear one was an inverted π shaped at 3 m from the stern and the Front one, an inverted T, 33 m from the stern. Both were selected made of Steel and with a Section Profile NACA 66-206, as

can be seen in Figures 16a, b and c. The steel roughness was chosen to be 0.15 mm.

In the Foil Geometry definition, Figures 13 and 25, the wingspan of the Rear foil is 10 m, with a strut span of 8.5 m (inverted π shape), while the Front foil's wingspan is 7.0 m and the strut span, zero (inverted T shape).

These inputs gave consistent feasible foil calculations, as the Rear and Front foils names in the top of the columns turned green. For the selected foil section, the program suggests and computes 95% of the stall angle (8 deg, see Figure 13 and 16) for the maximum angle of attack (α_{Tmax}), but this can be overwritten.

	Rear	Front
Foil Wingspan[b]	10	7
Strut Span [m]	8,5	0
α_{Tmax}	7,6	7,6

Figure 25 – Foil Geometry spans input

The optimized **Rear foil**, in Figures 17 and 18:

Chord = 2.765 m, $\alpha_T = 2.928$ deg, $W/S = 5239$ kgf/m²

The optimized **Front foil**, in Figures 26 and 27:

Chord = 1.893 m, $\alpha_T = 2.993$ deg, $W/S = 4961$ kgf/m²

Foil Geometry	Rear Foil Definition	Front Foil Definition	Foil Resistance	Resistance, Power and Operation	Formulation				
ChartTable									
Lift [N]	645216.22	W/S[kg/m²]	4800	4850	4900	4950	5000	5050	5100
Chord [m]	1.893	Area[m²]	13,696	13,554	13,416	13,281	13,148	13,018	12,89
Cl Alfa	0,075	Chord[m]	1,957	1,936	1,917	1,897	1,878	1,86	1,841
Alfa [Degree]	2.993	ImmStrut Ht[m]	4,057	4,036	4,017	3,997	3,978	3,96	3,941
W/S [kg/m²]	4961	Aspect Ratio	3,577	3,616	3,652	3,69	3,727	3,763	3,802
W/S Pratical Limits 3800-5700 kg/m²		Calkins Corr.	0,641	0,644	0,646	0,649	0,651	0,653	0,655
		Wadlin Corr.	0,986	0,986	0,986	0,986	0,986	0,987	0,987
		Corrected AR	3,577	3,616	3,652	3,69	3,727	3,763	3,802
		Cl alpha	0,074	0,074	0,075	0,075	0,075	0,075	0,076
		Cl required	0,217	0,219	0,222	0,224	0,226	0,228	0,231
		Ang. Att	2,932	2,959	2,96	2,987	3,013	3,04	3,039
		Cav. Lim. [t/c]	0,2351	0,2342	0,2333	0,2324	0,2315	0,2306	0,2298
		Stru. Lim. [t/c]	0,0571	0,058	0,0589	0,0598	0,0607	0,0616	0,0625
		CL/CD max[t/c]	0,06	0,06	0,06	0,06	0,06	0,06	0,06

Figure 26 – Front Foil Definition Table (where: Cav. Lim. – cavitation limit; Stru. Lim. – structural limit)

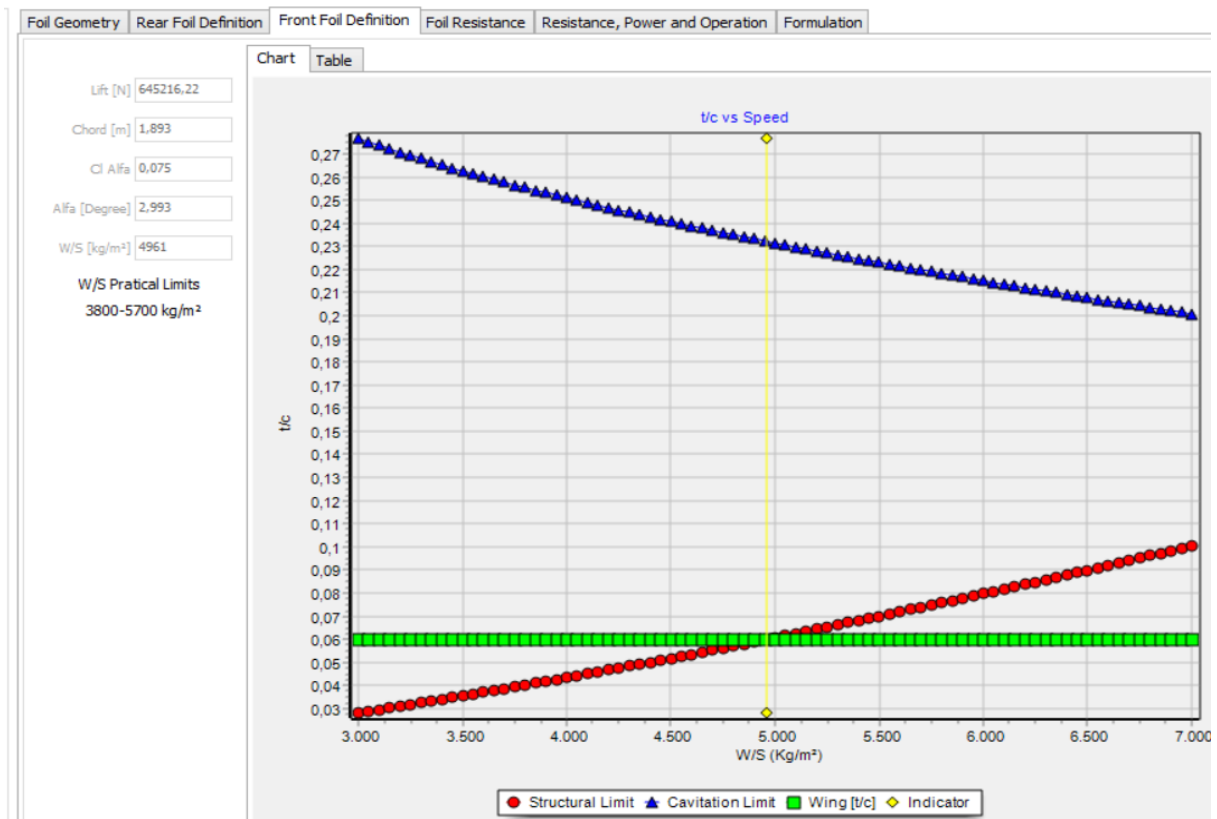


Figure 27 – Front Foil Optimization Chart

Soon after, the Take-off calculations take part in the Foil Resistance Area. The results of the calculations, as explained in Section 4, are shown in Figure 21, the option used being 'Compute Take-off from Initial Angle' and the resulting take-off speed is 25 kt. The take-off process is shown in Figure 19, where the Pre Take-off data is shown. In Figure 20 the decay of the Angle of Attack after the take-off can be seen, as well as the Foils' Lift and Drag vs Speed curves, up to, and a little beyond the cruise speed. The Cruising Condition (at 40 kt) is also shown.

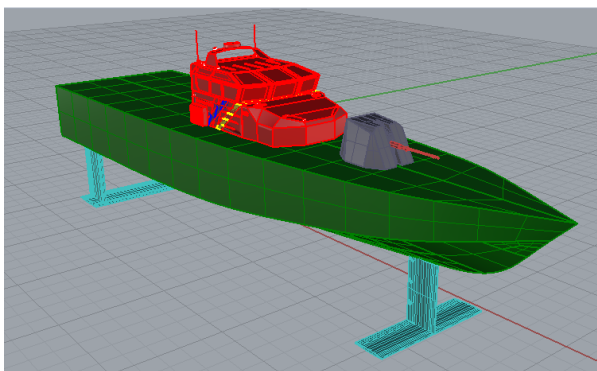


Figure 28 – Perspective drawing of the hydrofoil patrol boat conceptual design

Finally, the Resistance and Power Curves are drawn in the Resistance, Power and Operation Area. The

curves can be seen in Figure 23. The calculations are displayed in Figure 22. It should be noticed that until 14 kt the method applied is Holtrop (1984), and for 15 kt on, Savitsky (1964), in the *Planing Boat* program. The Resistance and Power vs Speed curves of the original planing boat, still without foils, is presented for comparison.

In Figure 28 a perspective view of the conceptual design of the hydrofoil patrol boat is presented. This was taken from the General Arrangement produced in the design.

7. Conclusion

The development of a computer program that dimensions hydrofoil boats has been presented. It is integrated to a full computer naval architectural design system, but can also be used autonomously. It is in a suite that calculates the Dynamic Equilibrium, Resistance and Propulsion for these craft and also planing boats, displacement and high-speed displacement boats, complimenting and making use of their calculations and results. It is the only software known, and therefore innovative, to optimize and determine, from the selection of foils section profiles, configurations and material, the dimensions of the foils, their full integration with the hull used, take-off speed

determination, lift and drag vs speed curves, angles of attack at each speed, resistance and power vs speed for propulsion. In other words, it doesn't only dimension the foils like it could be adapted for CFD software; it does it in a whole boat design integrated manner. In fact, since the traditional way of designing hydrofoil boats can get complex, the method of Schachter and Vianna (1996) used here is also a breakthrough.

The analytical empirical formulation encoded, well based in the literature, provides a good level of reliability (as the precision check and the design example showed) and added to the previous ones provides a unique, novel, set of tools for the hydrodynamic design of high-speed or high performance (as it is called today) craft.

The program's ergonomics, consistent with the other modules in EqDin and in the Solution-Focused Design System, based on the mentioned Solution-Focused Design philosophy, provide interfaces with menus easy to use, logical design sequence with different design factors that relate to each other in the same screen (interdisciplinary screens, e.g., with seakeeping, structures and geometry), with explicit units, placing inputs visually close to outputs, allowing for changes to evaluate alternatives along the design in real time, at any time, in any order, with inconsistency warning, with defaults for general settings, producing reports, all very quickly, which is rare in existing software in the market, that require either input files or complicated settings in different tabs. It also produces comprehensive graphics with tables showing all calculations of all parameters. It is believed this software may be useful, mainly as it is being noticed a return of this kind of craft in the market, for recreational and military boats.

8. Acknowledgements

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